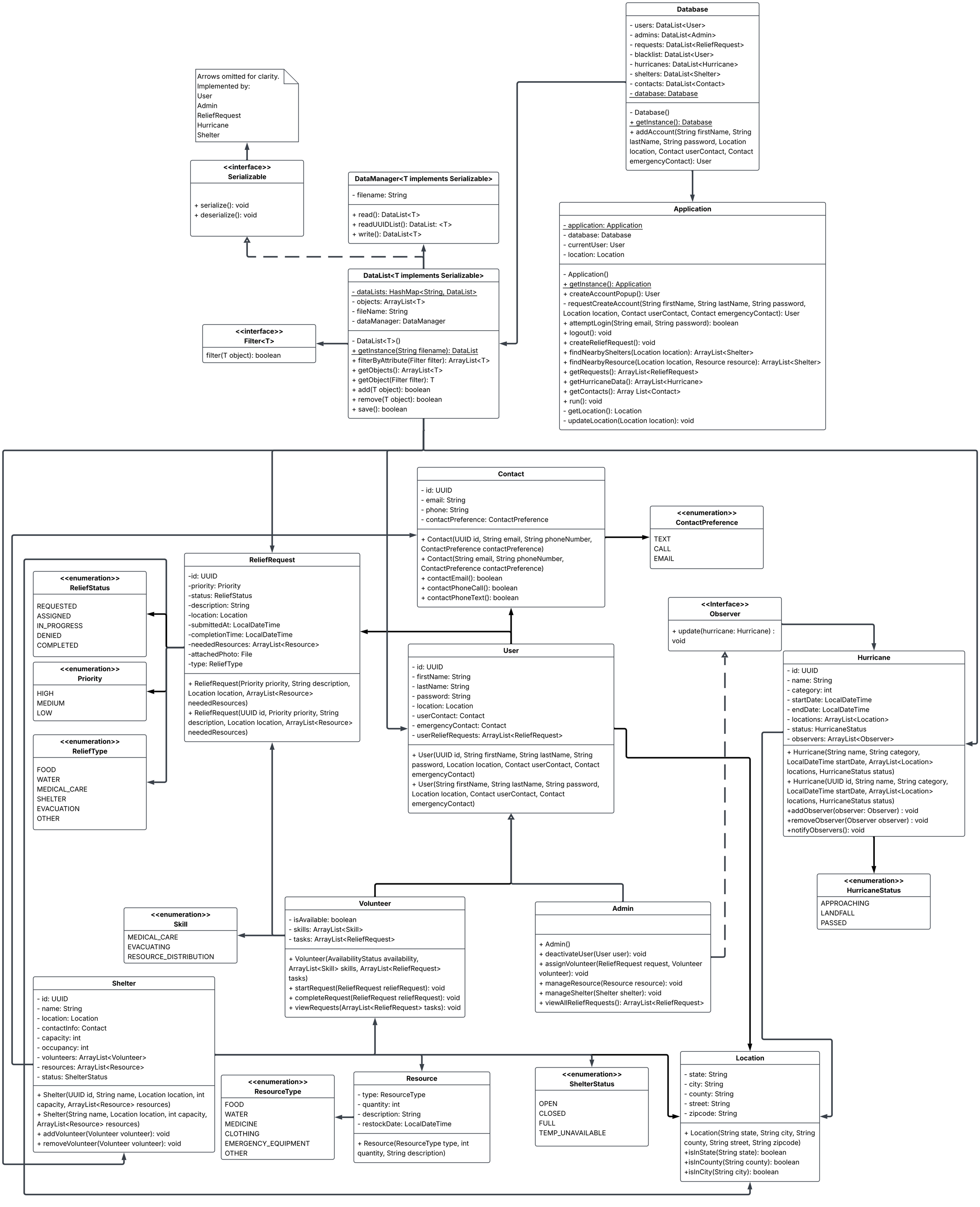

Arrows omitted for clarity.
Implemented by:
User
Admin
ReliefRequest
Hurricane
Shelter
<<interface>>
Serializable
+ serialize(): void
+ deserialize(): void
DataManager<T implements Serializable>
- filename: String
+ read(): DataList<T>
+ readUUIDList(): DataList: <T>
+ write(): DataList<T>
Database
- users: DataList<User>
- admins: DataList<Admin>
- requests: DataList<ReliefRequest>
- blacklist: DataList<User>
- hurricanes: DataList<Hurricane>
- shelters: DataList<Shelter>
- contacts: DataList<Contact>
- database: Database
- Database()
+ getInstance(): Database
+ addAccount(String firstName, String lastName, String password, Location location, Contact userContact, Contact emergencyContact): User
Application
- application: Application
- database: Database
- currentUser: User
- location: Location
- Application()
+ getInstance(): Application
+ createAccountPopup(): User
- requestCreateAccount(String firstName, String lastName, String password, Location location, Contact userContact, Contact emergencyContact): User
+ attemptLogin(String email, String password): boolean
+ logout(): void
+ createReliefRequest(): void
+ findNearbyShelters(Location location): ArrayList<Shelter>
+ findNearbyResource(Location location, Resource resource): ArrayList<Shelter>
+ getRequests(): ArrayList<ReliefRequest>
+ getHurricaneData(): ArrayList<Hurricane>
+ getContacts(): Array List<Contact>
+ run(): void
- getLocation(): Location
- updateLocation(Location location): void
DataList<T implements Serializable>
- dataLists: HashMap<String, DataList>
- objects: ArrayList<T>
- fileName: String
- dataManager: DataManager
- DataList<T>()
+ getInstance(String filename): DataList
+ filterByAttribute(Filter filter): ArrayList<T>
+ getObjects(): ArrayList<T>
+ getObject(Filter filter): T
+ add(T object): boolean
+ remove(T object): boolean
+ save(): boolean
<<interface>>
Filter<T>
filter(T object): boolean
Contact
- id: UUID
- email: String
- phone: String
- contactPreference: ContactPreference
+ Contact(UUID id, String email, String phoneNumber, ContactPreference contactPreference)
+ Contact(String email, String phoneNumber, ContactPreference contactPreference)
+ contactEmail(): boolean
+ contactPhoneCall(): boolean
+ contactPhoneText(): boolean
<<enumeration>>
ContactPreference
TEXT
CALL
EMAIL
ReliefRequest
-id: UUID
-priority: Priority
-status: ReliefStatus
-description: String
-location: Location
-submittedAt: LocalDateTime
-completionTime: LocalDateTime
-neededResources: ArrayList<Resource>
-attachedPhoto: File
-type: ReliefType
+ ReliefRequest(Priority priority, String description, Location location, ArrayList<Resource> neededResources)
+ ReliefRequest(UUID id, Priority priority, String description, Location location, ArrayList<Resource> neededResources)
<<enumeration>>
ReliefStatus
REQUESTED
ASSIGNED
IN_PROGRESS
DENIED
COMPLETED
<<enumeration>>
Priority
HIGH
MEDIUM
LOW
<<enumeration>>
ReliefType
FOOD
WATER
MEDICAL_CARE
SHELTER
EVACUATION
OTHER
User
- id: UUID
- firstName: String
- lastName: String
- password: String
- location: Location
- userContact: Contact
- emergencyContact: Contact
- userReliefRequests: ArrayList<ReliefRequest>
+ User(UUID id, String firstName, String lastName, String password, Location location, Contact userContact, Contact emergencyContact)
+ User(String firstName, String lastName, String password, Location location, Contact userContact, Contact emergencyContact)
<<Interface>>
Observer
+ update(hurricane: Hurricane) : void
Hurricane
- id: UUID
- name: String
- category: int
- startDate: LocalDateTime
- endDate: LocalDateTime
- locations: ArrayList<Location>
- status: HurricaneStatus
- observers: ArrayList<Observer>
+ Hurricane(String name, String category, LocalDateTime startDate, ArrayList<Location> locations, HurricaneStatus status)
+ Hurricane(UUID id, String name, String category, LocalDateTime startDate, ArrayList<Location> locations, HurricaneStatus status)
+addObserver(observer: Observer) : void
+removeObserver(Observer observer) : void
+notifyObservers(): void
<<enumeration>>
HurricaneStatus
APPROACHING
LANDFALL
PASSED
<<enumeration>>
Skill
MEDICAL_CARE
EVACUATING
RESOURCE_DISTRIBUTION
Volunteer
- isAvailable: boolean
- skills: ArrayList<Skill>
- tasks: ArrayList<ReliefRequest>
+ Volunteer(AvailabilityStatus availability, ArrayList<Skill> skills, ArrayList<ReliefRequest> tasks)
+ startRequest(ReliefRequest reliefRequest): void
+ completeRequest(ReliefRequest reliefRequest): void
+ viewRequests(ArrayList<ReliefRequest> tasks): void
Admin
+ Admin()
+ deactivateUser(User user): void
+ assignVolunteer(ReliefRequest request, Volunteer volunteer): void
+ manageResource(Resource resource): void
+ manageShelter(Shelter shelter): void
+ viewAllReliefRequests(): ArrayList<ReliefRequest>
Shelter
- id: UUID
- name: String
- location: Location
- contactInfo: Contact
- capacity: int
- occupancy: int
- volunteers: ArrayList<Volunteer>
- resources: ArrayList<Resource>
- status: ShelterStatus
+ Shelter(UUID id, String name, Location location, int capacity, ArrayList<Resource> resources)
+ Shelter(String name, Location location, int capacity, ArrayList<Resource> resources)
+ addVolunteer(Volunteer volunteer): void
+ removeVolunteer(Volunteer volunteer): void
<<enumeration>>
ResourceType
FOOD
WATER
MEDICINE
CLOTHING
EMERGENCY_EQUIPMENT
OTHER
Resource
- type: ResourceType
- quantity: int
- description: String
- restockDate: LocalDateTime
+ Resource(ResourceType type, int quantity, String description)
<<enumeration>>
ShelterStatus
OPEN
CLOSED
FULL
TEMP_UNAVAILABLE
Location
- state: String
- city: String
- county: String
- street: String
- zipcode: String
+ Location(String state, String city, String county, String street, String zipcode)
+isInState(String state): boolean
+isInCounty(String county): boolean
+isInCity(String city): boolean